UI / UX

Usability

The effectiveness with which users can accomplish tasks in a software system.

Learnability: Learnability means how easy it is for a new user to learn how to use the system.

Efficiency: Efficiency means how quickly users can perform tasks after they have learned the system.

Safety: safety means minimizing user errors and allowing users to recover from error

Concept	Meaning
Learnability	Easy to learn
Efficiency	Fast after learning
safety	Few errors + easy recovery

10 Usability Heuristics

- Visibility
- Mapping
- Freedom
- Consistency
- Error prevention
- Recognition
- Flexibility
- Minimalism
- Error recovery
- Help

Steps in UI Design

Planning → Prototyping and sketching →

Optimization → Implement

Planning

- understand customer values
- Brainstorming and research
- Make Feature List
- Plan the navigation

ketching and Prototyping

- Wireframe modeling

: Wireframe হচ্ছে UI-এর basic structure/layout

- Prototyping can be done
  - Manually (paper-pencil, pen tablet)
  - Electronically (Adobe XD, Photoshop, Figma)

Benefits of Prototyping

- Prototype থাকলে developer বুঝতে পারে final UI কীভাবে তৈরি করতে হবে।
- Prototype বানানোর সময় missing requirements বের হয়ে আসতে পারে
- Prototype দেখে সম্ভাব্য user interactions আগে থেকে identify করা যায়।
- Prototype user কে দেখিয়ে feedback নেওয়া যায় final software তৈরি করার আগেই।

What are the best practices for achieving usability?

- User testing and field studies
- UI expert evaluation/review
- Avoid being overwhelmed by technicalities

Two Types of User Interface

Graphical User Interface (GUI)

A GUI allows users to interact with an operating system or software using graphical elements.

Command Line User Interface (CLI)

CLI allows users to interact with an operating system by typing commands in a terminal or console.

Scheinderman's 8 Golden Rules

1. Strive for consistency

- একই ধরনের কাজের জন্য একই ধরনের design, terminology, command এবং behavior ব্যবহার করতে হবে।

2. Enable frequent users to use shortcuts

- Experienced users একই কাজ বারবার করলে তারা কম interaction - এ দ্রুত কাজ করতে চায়। তাই shortcuts দিতে হবে।

Examples:

ctrl+s → save

ctrl+c → copy

ctrl+v → paste

3. Offer Informative Feedback

User কোনো Action করলে system এর পক্ষ থেকে feedback থাকা উচিত।

Example: যদি file upload করতেন:

Good UI: Uploading 70%

4.) Design Dialog to yield closure.

ঃ প্রতিটা action এর sequence মনে চিন্তা

Ex:

Step 1 → Personal Info

Step 2 → Academic Info

Step 3 → Payment

Step 4 → Confirmation

শেষে:

Registration completed successfully!

এটি closure.

5. Offer simple Error Handling

- System এমনভাবে design করা উচিত যেন :

(i) Serious error prevent করা যায়

(ii) Error detect করা যায়

(iii) User-কে সহজভাবে explain করা যায়

(iv) Recovery mechanism রাখা যায়

Error 0x87 3962 [Not clear error]

Invalid email address, please enter a valid email.

[clear error]

6. Permit easy Reversal of actions

- User যেন ভুল করলে undo/cancel/back করতে পারে

7. Let the User Be in Control

- User should control the system, not the system control the user.

Example: যুদি file copy করতেছে

Button: Cancel copy

User চাইলে operation stop করতে পারে।

Scenario: An application automatically starts a process and does not allow the user to stop or cancel it.

Rule 8. Reduce short Term memory Load

মানুষের short-term memory limited.

Scenario: A user must remember information from a previous screen before completing the current task.